

Sandro Loch

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Summary

Backend, DevOps, and Data Platform Engineer with 5+ years of experience building and operating data-intensive systems, backend services, and automated delivery pipelines. Strong background in Python, Django, PostgreSQL, Docker, CI/CD, and Elasticsearch, with experience in production data workflows, API integration, infrastructure automation, and interactive data visualization. Contributed to Plotly chart modernization in AlertaDengue/InfoDengue and to AI-powered RAG pipelines for large-scale scientific and legal document processing. Focused on reliability, maintainability, and operational efficiency.

Experience (last 5 years)

DevOps & Data Platform Engineer, AlertaDengue / InfoDengue (FIOTEC) Nov 2025 – Present

- Lead reliability, automation, and operational stability for the InfoDengue and AlertaDengue platforms, including monitoring and maintenance of pipelines integrating SINAN and meteorological data.
- Manage CI/CD workflows, dependency updates, and test-backed automation to reduce manual operations and improve release safety.
- Maintain processing servers, Linux environments, Python runtimes, and database output layers supporting the web platform and integration services.

AI-Powered Full-Stack & DevOps Engineer, OSL - Open Science Labs / LiteRev - Mar 2023 – Oct 2025
Universite de Geneve

- Designed and operated backend infrastructure with Docker, NGINX, Django, Celery, Redis, and Elasticsearch for large-scale academic and legal document processing.
- Developed RAG workflows using LLMs, vector search, and contextual retrieval to improve scientific literature analysis.
- Implemented CI/CD pipelines in GitHub Actions to automate testing, builds, and deployments across backend services.
- Built ingestion and normalization pipelines for arXiv, MedRxiv, and PubMed Central using Python and Django REST Framework.

Tech Lead - SDX & MyMHAI, TeleHealthCareAI initiative Oct 2024 – Dec 2025

- Coordinated sprint planning and weekly cross-functional follow-ups across development, clinical research, and operations.
- Structured issue tracking and milestone management in GitHub Projects to improve planning and execution.

Full-Stack Backend Developer, Foundation for Scientific and Technological Aug 2019 – Apr 2024
Development in Health (FIOTEC)

- Developed Django applications and REST APIs for dengue and arbovirus surveillance, backed by PostgreSQL data models and optimized queries.
- Automated development and delivery workflows using Ansible, Docker, Celery, PostgreSQL, and CI/CD tooling.
- Processed large epidemiological datasets with Pandas and Dask and produced interactive analytical outputs.
- Migrated and modernized AlertaDengue/InfoDengue charts to Plotly, including homepage and reporting views used in production.
- Automated climate data ingestion and supported high availability with PostgreSQL replication, validation scripts, and MinIO-based storage services.

Open Source Contributions

- **EpiGraphHub (TheGraphNetwork)**: Contributed to epidemiological data platforms for public health analysis, with emphasis on data integration.
- **Open Science Labs**: Contributed to open-source tools focused on data accessibility, reproducibility, and scientific research workflows.

Selected Certifications

- **Building Agentic RAG with LlamaIndex** - deeplearning.ai (Jun 2025)
- **MySQL Explorer** - Oracle University (Sep 2024)
- **Scrum Foundation Professional Certification (SFPC)** - CertiProf (Jul 2023)
- **Data Analysis for Health Surveillance** - UFSC (May 2023)
- **PostgreSQL DBMS** - IFRS (Apr 2022)
- **Statistical Foundations** - LabTime / CAPES (Nov 2020)
- **Introduction to Data Visualization with Matplotlib** - DataCamp (Sep 2020)
- **Unit Testing for Data Science in Python**; DataCamp (Jun 10,2020)
- **Bash Scripting**; DataCamp (May 20, 2020)
- **Manipulating DataFrames with Pandas** - DataCamp (Jan 2020)

Education

University Leonardo Da Vinci, Santa Catarina, Brazil , Bachelor's Degree in Big Data and Analytical Intelligence	Sept 2021 – Nov 2024
WorldQuant University (WQU), New Orleans, Louisiana, USA , Data Science Lab	Dec 2024 – In Progress